Test Case ID	Target Function / Operation	Test Input Vector	Expected Output	Actual Output — Legacy Code	Pass / Fail	Technical Remarks
TC-01	add_item()	Add 10, 20, 30	All three values are stored successfully. currentCount = 3, maxCapacity = 5.	Items are added successfully.	**PASS**	Normal insertion works while the initial capacity has not been exceeded.
TC-02	add_item() — Dynamic Resize	Add 10, 20, 30, 40, 50, 60	Capacity should increase from 5 to 10; all six values should remain accessible.	Six items are added and capacity increases, but the original array is not released.	**FAIL**	Dynamic resizing works functionally but causes a heap memory leak because delete[] dataArray is missing.

TC-03	printAll()	Add 10, 20, 30, 40, 50, then display.	Output should contain exactly 10 20 30 40 50.	Attempts to print five valid elements plus one additional value beyond the valid range.	**FAIL**	i <= currentCount causes an off-by-one heap read. Correct condition should be i < currentCount.
TC-04	findItem() — Existing Value	Search for 30 in {10,20,30,40,50}.	Returns index 2.	Returns 2.	**PASS**	Linear search successfully finds an existing element. The redundant inner loop is unnecessary.
TC-05	findItem() — Missing Value	Search for 999 in {10,20,30,40,50}.	Returns -1.	Returns -1.	**PASS**	Correctly identifies that the target does not exist.

TC-06	remove_item_at() — Valid Index	Remove index 2 from {10,20,30,40,50}.	30 is removed; remaining array becomes {10,20,40,50} and currentCount = 4.	Element is removed and remaining elements shift left correctly.	**PASS**	Functionally correct for a valid index, although the nested loop is unnecessarily complicated.
TC-07	remove_item_at() — Negative Index	Remove index -5.	Operation should be rejected and array state should remain unchanged.	Prints "Invalid index!"; array remains unchanged.	**PASS**	Negative-index validation is implemented correctly.
TC-08	remove_item_at() — Upper Out-of-Bounds	Remove index 99 when currentCount = 4.	Operation should be rejected safely; no memory should be accessed or modified.	No upper-bound validation exists; function may access memory outside the valid array range and cause undefined behavior.	**FAIL**	Critical defect. The function checks idx < 0 but does not check idx >= currentCount.

TC-09	processMatrix()	Transpose {{1,2,3},{4,5,6},{7,8,9}}.	Expected result: {{1,4,7},{2,5,8},{3,6,9}}.	Correct transposed matrix is displayed.	**PASS**	Matrix transposition logic is functionally correct for the hardcoded 3 × 3 matrix.